BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Which one of the following is a lewis base?

(A) HCl
(B) $AlCl_3$
(C) BF_3
(D) F^-

Q2. The units of Kc for the reaction $PCl_3 + Cl_2 \rightleftharpoons PCl_5$ is.

(A) $mol.dm^{-3}$
(B) $mol^{-1}.dm^{-3}$
(C) $mol^{-1}.dm^3$
(D) $mole.dm^3$

Q3. Salts formed when OH^- ions of a base are partially neutralized by an acid are called.

(A) Basic salts
(B) Acidic salts
(C) Neutral salts
(D) All of these

Q4. Natural gas is mainly composed of.

(A) CH_4
(B) C_2H_8
(C) C_2H_6
(D) C_2H_2

Q5. A group of atoms that is formed when one hydrogen atom is removed from an alkane molecule is called.

(A) Functional group
(B) Alkyl radical
(C) Alkane
(D) Alkyne

Q6. Substitution reaction is the characteristic of.

(A) Alkynes
(B) Alkenes
(C) Alkanes
(D) None of these

Q7. Triglycerides are the building blocks of.

(A) Proteins
(B) Carbohydrates
(C) Lipids
(D) Vitamins

Q8. The coldest region in the atmosphere is.

(A) Troposphere
(B) Stratosphere
(C) Mesosphere
(D) Thermosphere

Q9. The density of water is maximum at.

(A) -40°C
(B) 40°C
(C) -4°C
(D) 4°C

Q10. All steps are used for the extraction of metals except.

(A) Roasting
(B) Smelting
(C) Evaporation
(D) Bessmirization

Q11. Permanent hardness of water is removed by adding.

(A) Washing soda
(B) Soda lime
(C) Caustic soda
(D) Baking soda

Q12. Which one of the following is a disaccharide?

(A) Glucose
(B) Fructose
(C) Sucrose
(D) Starch

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Differenciate between forward and Reverse reactions & give one example of each.

(ii) Distinguish strong acids from weak acids. Give two examples of each.

(iii) Calculate the P^{H} and P^{OH} of 0.5 M solution of HCl?

(iv) Write any five uses of organic compounds?

(v) What happens when alkyl halide is reduced?

(vi) Distinguish between Fats and oils?

(vii) What is the composition of atmosphere?

(viii) Differenciate between soft and hard water?

(ix) Define Metallurgy? What are its steps.

(x) Write any five uses of carbohydrates.

(xi) Discuss how the acid rain affects the aquatic life?

SECTION "C"

Maximum Marks: 21

> Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] Derive an expression for equilibrium constant for a general reaction.

(b) [3] Differenciate between Acidic salt and neutral salt. Give two examples of each.

QIV.

(a) [3] Classify the organic compound as saturated and unsaturated? Give examples of each.

(b) [4] Write two methods for the preparation of Alkenes?

QV.

(a) [4] Define the term vitamin and classify it?

(b) [3] Differenciate between primary and secondary pollutants.

QVI.

(a) [4] Write the different uses of Urea?

(b) [3] Write the chemical reaction of water with i) C ii) Cl_2 iii) CaO